

Fundamental, Computational, and Conceptual Questions in Probability and Bayesian Inference

Fundamental, Computational, and Conceptual Questions and Answers

1. What is a random variable?

A random variable is a variable that can take on different values, each with an associated probability, depending on the outcome of a random process.

2. State the axioms of probability.

- $0 \leq P(A) \leq 1$
- $P(\text{True}) = 1, P(\text{False}) = 0$
- $P(A \vee B) = P(A) + P(B) - P(A \wedge B)$

3. What is joint probability?

Joint probability is the probability of two (or more) events happening together, e.g., $P(A \wedge B)$.

4. Explain marginal probability.

Marginal probability is the probability of an event irrespective of the outcome of other variables, found by summing joint probabilities over the unwanted variables.

5. What is conditional probability?

Conditional probability $P(A|B)$ is the probability of event A occurring given that B has occurred, calculated as $P(A|B) = \frac{P(A \wedge B)}{P(B)}$.

6. What does it mean for two events to be independent?

Two events are independent if the occurrence of one does not affect the probability of the other, i.e., $P(A \wedge B) = P(A)P(B)$.

7. What is conditional independence?

Events A and B are conditionally independent given C if $P(A \wedge B|C) = P(A|C)P(B|C)$.

8. State the product rule of probability.

$P(A, B) = P(A|B)P(B) = P(B|A)P(A)$

9. **State the sum rule of probability.**

$$P(A \vee B) = P(A) + P(B) - P(A, B)$$

10. **What is the law of total probability?**

If $A_1, \dots, A_n$ are mutually exclusive and exhaustive, $P(B) = \sum_i P(B|A_i)P(A_i)$.

11. **State Bayes' theorem.**

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

12. **Interpret prior, likelihood, and posterior in Bayes' theorem.**

- Prior: $P(A)$, initial belief before new evidence.
- Likelihood: $P(B|A)$, probability of evidence given the hypothesis.
- Posterior: $P(A|B)$, updated belief after seeing evidence.

13. **What is the marginal probability in Bayes' theorem?**

The denominator $P(B)$, calculated as $\sum_i P(B|A_i)P(A_i)$, ensures the posterior is a valid probability.

14. **Give an example of applying Bayes' theorem in real life.**

Medical diagnosis: Given a positive test result, Bayes' theorem updates the probability of having a disease based on test accuracy and disease prevalence.

15. **How does Bayes' theorem relate to machine learning?**

It provides a framework for updating model beliefs as new data becomes available, underpinning probabilistic models and classifiers like Naive Bayes.

16. **Compute $P(A|B)$ if $P(A) = 0.3$, $P(B|A) = 0.8$, $P(B) = 0.5$.**

$$P(A|B) = \frac{0.8 \times 0.3}{0.5} = 0.48$$

17. **Given $P(D) = 0.01$, $P(\text{Pos}|D) = 0.95$, $P(\text{Pos}|\sim D) = 0.05$, what is $P(D|\text{Pos})$?**

$$P(D|\text{Pos}) = \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.05 \times 0.99} \approx 0.161$$

18. **A factory's 3 machines A, B, C produce 25%, 35%, 40% of bolts with defect rates 5%, 4%, 2%. A bolt is defective. Probability it came from B?**

$$P(B|\text{Def}) = \frac{0.35 \times 0.04}{0.25 \times 0.05 + 0.35 \times 0.04 + 0.4 \times 0.02} \approx 0.304$$

19. **If a coin is flipped 10 times and lands heads 7 times, what is the MLE for probability of heads?**

$$\text{MLE: } p = 7/10 = 0.7$$

20. **If you have a prior belief that $p = 0.5$ (Beta(2,2)), and observe 7 heads in 10 flips, what is the MAP estimate?**

$$\text{MAP: } \theta^* = \frac{7+2-1}{10+2+2-2} = \frac{8}{12} = 0.667$$

21. **What is the difference between MLE and MAP estimation?**

MLE estimates parameters by maximizing likelihood from data alone; MAP incorporates prior beliefs, maximizing the posterior.

22. **When does MAP estimation become equivalent to MLE?**
When the prior is uniform (uninformative), MAP and MLE give the same result.
23. **Why is the Naive Bayes classifier called “naive”?**
Because it assumes all features are conditionally independent given the class, which is rarely true in practice.
24. **What is the main computational advantage of Naive Bayes?**
The independence assumption allows the joint likelihood to be decomposed into a product of individual likelihoods, simplifying calculations.
25. **What is a conjugate prior?**
A prior distribution that, when combined with the likelihood, yields a posterior in the same family, making Bayesian updating analytically tractable.
26. **What is the posterior predictive distribution?**
The distribution of future observations, integrating over the posterior of model parameters.
27. **Explain the Monty Hall problem in terms of conditional probability.**
After the host opens a door with a goat, switching increases your chance of winning from $1/3$ to $2/3$, as conditional probabilities update based on new information.
28. **What is Bayesian model averaging?**
Combining predictions from multiple models, weighted by their posterior probabilities, to account for model uncertainty.
29. **What is the role of the likelihood function in Bayesian inference?**
It quantifies how probable the observed data is under different parameter values, shaping the posterior.
30. **What is the difference between prior and posterior?**
The prior is the belief before seeing data; the posterior is the updated belief after incorporating data.
31. **What is the purpose of the marginal likelihood (evidence) in Bayesian inference?**
It normalizes the posterior and is used for model comparison.
32. **How is Bayesian inference used in decision making?**
By updating probabilities as evidence accumulates, it supports rational decision-making under uncertainty.
33. **What is Markov Chain Monte Carlo (MCMC)?**
A computational method to sample from complex posteriors when analytical solutions are intractable.

34. **What is variational inference?**
An optimization-based technique to approximate complex posteriors with simpler distributions.
35. **How does Bayesian learning differ from frequentist learning?**
Bayesian learning incorporates prior knowledge and updates beliefs; frequentist relies solely on observed data.
36. **What is the maximum a posteriori (MAP) estimate for a Gaussian likelihood with a Gaussian prior?**
The MAP estimate is a weighted average of the prior mean and the sample mean, weighted by their variances.
37. **How does the Naive Bayes classifier handle continuous features?**
By assuming a parametric distribution (e.g., Gaussian) for each feature, estimating parameters from data.
38. **Why is the denominator in Bayes' theorem often ignored in classification?**
Because it is the same for all classes, so only the numerator matters for comparing class probabilities.
39. **What is overfitting in the context of Bayesian inference?**
Overfitting occurs when the model is too flexible, fitting noise in the data; Bayesian priors can regularize and prevent this.
40. **What is the predictive distribution in Bayesian regression?**
It is the distribution over possible outcomes for a new input, integrating over the posterior of the regression parameters.
41. **What is the fundamental principle behind Bayesian algorithms?**
Incorporating prior knowledge.
42. **Which Bayesian algorithm is commonly used for classification?**
Naive Bayes.
43. **Which assumption does Naive Bayes make about features?**
They are conditionally independent.
44. **Which algorithm is used for updating beliefs in Bayesian inference?**
Bayes' Theorem.
45. **In Bayesian inference, what does the posterior distribution represent?**
The updated belief after incorporating new evidence.
46. **Which algorithm estimates hyperparameters in Bayesian models?**
Markov Chain Monte Carlo (MCMC).
47. **How are models combined in Bayesian model averaging?**
By weighting their predictions based on their posterior probabilities.

48. **Which algorithm estimates parameters of a Bayesian neural network?**
Variational Inference.
49. **What is the purpose of model comparison in Bayesian model selection?**
To select models that best explain the data, considering both fit and complexity.
50. **What is the main advantage of Bayesian approaches in machine learning?**
They provide a principled way to incorporate prior knowledge and quantify uncertainty in predictions.